

DecisionPilot OS

Intelligent Next Best Action Platform

Technical Documentation

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1. Project Overview

Introduction

DecisionPilot OS is a reusable Agentic Decision Intelligence Platform designed to assist enterprises in making faster, more accurate, and explainable business decisions. The platform transforms customer interactions and enterprise knowledge into intelligent recommendations by orchestrating multiple specialized AI agents through a dynamic planner-based workflow.

Traditional business decision-making often depends on manual document analysis, fragmented customer information, and subjective human judgment. These approaches are time-consuming, inconsistent, and difficult to scale across large organizations. DecisionPilot OS addresses these challenges by automatically extracting enterprise knowledge, analyzing customer context, evaluating business risks, identifying opportunities, and recommending the next best actions with complete reasoning and supporting evidence.

The platform is designed as a reusable enterprise AI framework rather than a single-purpose chatbot. Its modular architecture allows organizations from different business domains to reuse the same platform by configuring business rules, workflows, knowledge sources, and specialized AI agents without changing the overall system architecture.

Business Domain

This implementation demonstrates the platform using a **B2B Enterprise SaaS Sales** use case, where organizations evaluate customer proposals, qualification criteria, business risks, and strategic opportunities before making sales and investment decisions.

Platform Objectives

The primary objectives of DecisionPilot OS are:

- Automate enterprise customer analysis using multiple AI agents.
- Retrieve and organize enterprise knowledge from uploaded customer documents.
- Perform customer discovery, ICP qualification, and business risk assessment.
- Generate intelligent next best action recommendations.
- Provide explainable AI decisions with supporting evidence and confidence scores.
- Enable Human-in-the-Loop review before business recommendations are accepted.
- Maintain reusable workflows and shared memory for future enterprise interactions.
- Deliver an intuitive dashboard for enterprise decision makers.

Key Capabilities

DecisionPilot OS provides the following capabilities:

- Dynamic Planner-Based Agent Orchestration
- Enterprise Knowledge Extraction
- Customer Discovery Intelligence
- Ideal Customer Profile (ICP) Qualification
- Enterprise Risk Assessment
- AI Decision Intelligence
- Recommendation Engine
- Explainable AI
- Interactive Enterprise Knowledge Graph
- AI Persona Intelligence
- Enterprise Analytics Dashboard
- Human-in-the-Loop Decision Review
- Shared Enterprise Memory

The platform is designed to be extensible, allowing additional agents, business rules, enterprise knowledge sources, and workflows to be integrated with minimal architectural changes.

2. Problem Statement

Business Challenge

Enterprise organizations receive large volumes of customer interactions through proposals, meeting notes, emails, CRM updates, product discussions, and business documents. These interactions contain valuable information that influences customer qualification, business opportunities, investment decisions, and strategic planning.

In many organizations, this information is analyzed manually by sales teams, solution architects, managers, and executives. The process is often slow, inconsistent, and dependent on individual experience. As customer data grows, it becomes increasingly difficult to identify important business signals, evaluate risks, and determine the most appropriate next actions.

Organizations also face challenges in consolidating information from multiple sources into a single decision-making process. Customer objectives, business priorities, risks, compliance

requirements, and organizational knowledge are frequently scattered across different documents and systems, making enterprise decision-making inefficient and error-prone.

Modern enterprises require an intelligent platform that not only retrieves relevant information but also understands business context, performs structured reasoning, explains its recommendations, and supports human decision-makers before actions are executed.

Problem Statement

How can organizations transform enterprise customer interactions and organizational knowledge into intelligent, explainable, and reusable business recommendations using a dynamic multi-agent AI platform?

Proposed Solution

DecisionPilot OS addresses this challenge by providing a reusable Agentic Decision Intelligence Platform that orchestrates multiple specialized AI agents to automate enterprise decision-making.

The platform ingests customer documents, extracts enterprise knowledge, performs customer discovery, evaluates customer qualification and business risks, generates explainable next best action recommendations, constructs an enterprise knowledge graph, supports Human-in-the-Loop review, and presents the results through an interactive analytics dashboard.

By combining planner-based orchestration, reusable AI agents, enterprise memory, explainable AI, and business reasoning, DecisionPilot OS enables organizations to make faster, more consistent, and data-driven business decisions while maintaining transparency and human oversight.

3. Solution Overview

Platform Overview

DecisionPilot OS is a reusable Agentic Decision Intelligence Platform designed to automate enterprise decision-making by coordinating multiple specialized AI agents through a dynamic planner-based workflow.

Instead of relying on a single AI model, the platform decomposes complex business problems into smaller tasks that are executed by dedicated agents. Each agent is responsible for a specific business capability such as enterprise knowledge extraction, customer discovery, customer qualification, risk assessment, decision intelligence, recommendation generation, explainability, analytics, and memory management.

The platform follows a modular architecture where every agent operates independently while sharing contextual information through a centralized workflow context. This design enables the platform to be reused across multiple enterprise domains by replacing business rules, knowledge sources, or specialized agents without modifying the overall architecture.

End-to-End Processing Flow

The platform processes enterprise customer documents through the following workflow:

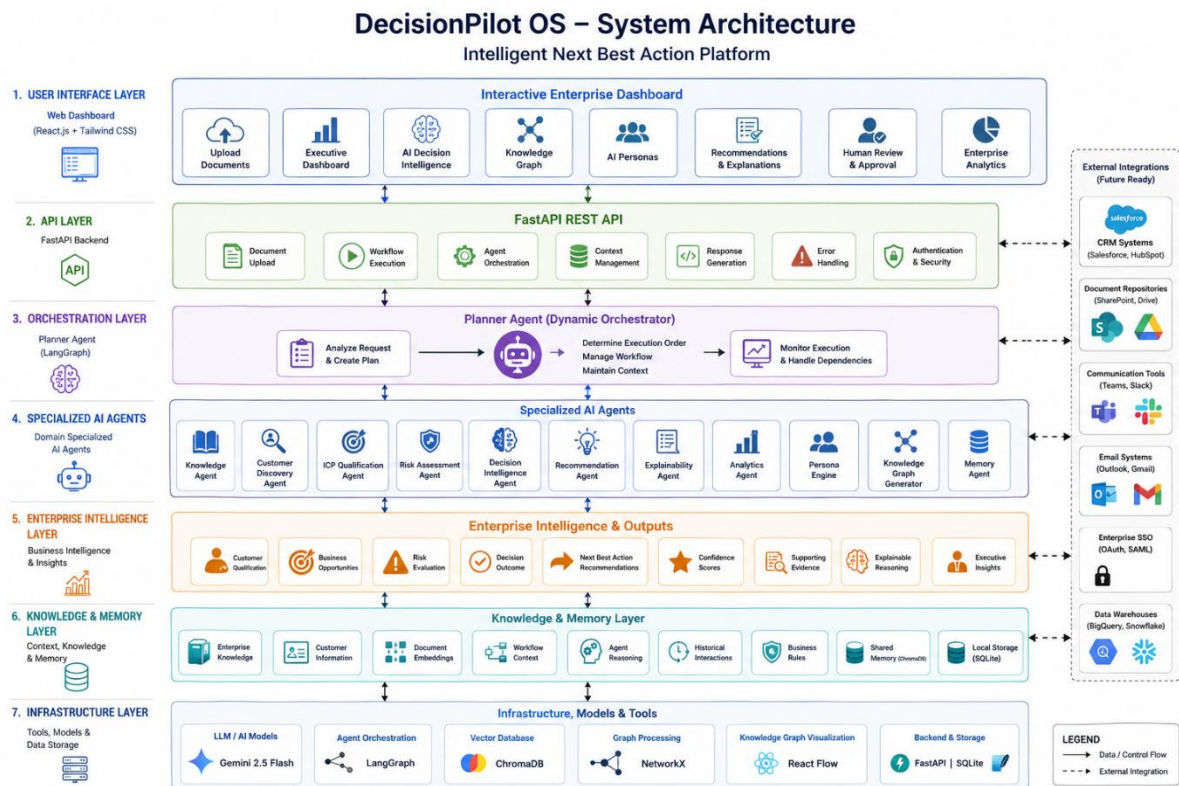
1. Customer documents are uploaded through the web interface.
2. The Planner Agent analyzes the request and creates an execution plan.
3. Specialized AI agents execute their assigned tasks sequentially while sharing context.
4. Enterprise knowledge is extracted and organized into structured business information.
5. Customer qualification, business risks, and strategic opportunities are evaluated.
6. AI generates explainable next best action recommendations with confidence scores.
7. Human users review the recommendations before business actions are accepted.
8. Results are presented through an interactive enterprise dashboard containing analytics, knowledge graphs, personas, recommendations, and explainable decision intelligence.

Architectural Design Principles

DecisionPilot OS has been designed around the following principles:

- **Modularity:** Each AI agent performs a single well-defined responsibility.
- **Reusability:** Business workflows can be adapted to different enterprise domains without redesigning the platform.
- **Extensibility:** New agents, business rules, and enterprise knowledge sources can be integrated with minimal effort.
- **Explainability:** Every recommendation includes supporting evidence, confidence scores, and reasoning to improve transparency.
- **Human-Centric Decision Making:** AI assists decision makers while Human-in-the-Loop review ensures governance and accountability.
- **Scalability:** The architecture supports future integration with enterprise systems such as CRM platforms, document repositories, collaboration tools, and knowledge management systems.

The result is a flexible and reusable enterprise AI platform capable of supporting intelligent decision-making across a wide range of business use cases.



4. System Architecture

Architecture Overview

DecisionPilot OS follows a modular Agentic AI architecture where a Planner Agent dynamically orchestrates multiple specialized AI agents to execute an enterprise decision-making workflow. Each agent performs a dedicated responsibility while sharing contextual information through a centralized workflow / Memory context. This approach improves modularity, reusability, explainability, and scalability.

The architecture separates user interaction, orchestration, business intelligence, knowledge processing, reasoning, analytics, and visualization into independent components. This enables new business workflows and AI agents to be integrated without changing the overall platform architecture.

Architecture Components

1. User Interface Layer

The User Interface provides the primary interaction point for enterprise users. Users upload customer documents, review AI-generated insights, visualize enterprise knowledge, inspect

recommendations, and perform Human-in-the-Loop decision review through an interactive dashboard.

Responsibilities

- Upload enterprise customer documents
 - Display analytics and business insights
 - Visualize enterprise knowledge graph
 - Display AI personas
 - Present recommendations and explanations
 - Support Human-in-the-Loop review
-

2. API Layer

The FastAPI backend exposes REST APIs that receive uploaded documents, initiate workflow execution, manage communication between the frontend and backend, and return structured results to the dashboard.

Responsibilities

- Document upload
 - Workflow execution
 - Agent coordination
 - API response generation
 - Error handling
-

3. Planner Agent

The Planner Agent acts as the central orchestrator of the platform. It dynamically determines the execution sequence of specialized agents based on the workflow requirements and ensures that contextual information is shared throughout the execution pipeline.

Responsibilities

- Create execution plan
- Orchestrate AI agents
- Manage workflow execution

- Maintain execution context
-

4. Specialized AI Agents

The platform contains multiple specialized AI agents, each responsible for a specific business capability.

The agents collaborate to perform:

- Enterprise knowledge extraction
- Customer discovery
- ICP qualification
- Business risk assessment
- Decision intelligence
- Recommendation generation
- Explainable AI
- Analytics generation
- Persona generation
- Knowledge graph construction
- Enterprise memory management

Each agent is independently reusable and can be replaced or extended without affecting other platform components.

5. Enterprise Intelligence Layer

The Enterprise Intelligence Layer combines outputs from multiple AI agents to generate business intelligence for enterprise decision makers.

This layer provides:

- Customer qualification
- Business opportunity analysis
- Risk evaluation
- Next Best Action recommendations
- Explainable AI reasoning

- Executive insights
-

6. Knowledge and Memory Layer

The platform maintains enterprise context using structured knowledge extraction and shared memory.

This layer stores:

- Enterprise knowledge
- Customer information
- Workflow context
- AI reasoning
- Historical interactions
- Business intelligence

Shared memory enables all AI agents to access consistent contextual information throughout workflow execution.

7. Visualization Layer

The final layer presents business intelligence through an interactive enterprise dashboard.

Visualizations include:

- Executive dashboard
- AI Decision Intelligence
- Customer Discovery
- Enterprise Knowledge Graph
- Risk Assessment
- AI Personas
- Explainable AI
- Recommendations
- Enterprise Analytics

This architecture enables DecisionPilot OS to function as a reusable Agentic Decision Intelligence Platform capable of supporting multiple enterprise business domains while maintaining scalability, modularity, and explainability.

5. Multi-Agent Architecture

Overview

DecisionPilot OS is built using a modular multi-agent architecture in which each AI agent is responsible for a single business capability. Instead of relying on one large AI model, the platform decomposes enterprise decision-making into specialized tasks executed by dedicated agents.

All agents are orchestrated by the Planner Agent and communicate through a shared Workflow Context. This architecture improves modularity, explainability, scalability, and reusability while allowing new agents to be added with minimal changes to the overall platform.

Agent Responsibilities

1. Planner Agent

Purpose

Acts as the central orchestrator of the platform.

Responsibilities

- Receives workflow requests
- Creates the execution plan
- Determines agent execution order
- Shares workflow context
- Coordinates the complete decision pipeline

Input

- Customer request
- Uploaded document

Output

- Execution workflow
- Shared context for downstream agents

2. Knowledge Agent

Purpose

Extracts structured enterprise knowledge from uploaded customer documents.

Responsibilities

- Extract business entities
- Organize enterprise information
- Build structured knowledge package

Output

- Enterprise Knowledge Package
-

3. Customer Discovery Agent

Purpose

Identifies important customer information required for business evaluation.

Responsibilities

- Identify customer goals
- Detect buying signals
- Extract business requirements
- Identify decision makers
- Estimate budget and timeline

Output

- Customer Discovery Profile
-

4. ICP Qualification Agent

Purpose

Evaluates whether the customer matches the organization's Ideal Customer Profile (ICP).

Responsibilities

- Calculate ICP score

- Evaluate customer fit
- Identify qualification level

Output

- ICP Score
 - Qualification Assessment
-

5. Risk Assessment Agent**Purpose**

Evaluates enterprise business risks using configurable business rules.

Responsibilities

- Execute business rules
- Identify operational risks
- Evaluate financial and compliance risks
- Generate enterprise risk score

Output

- Risk Assessment
 - Risk Scores
 - Triggered Rules
-

6. Decision Intelligence Agent**Purpose**

Generates the enterprise decision using outputs from previous agents.

Responsibilities

- Analyze customer qualification
- Evaluate business context
- Generate confidence score
- Produce enterprise decision

Output

- AI Decision
 - Confidence Score
-

7. Recommendation Agent

Purpose

Generates actionable Next Best Action recommendations.

Responsibilities

- Recommend business actions
- Prioritize next steps
- Support enterprise users

Output

- Recommended Actions
-

8. Explainability Agent

Purpose

Provides transparency for every AI-generated recommendation.

Responsibilities

- Generate AI reasoning
- Display supporting evidence
- Explain confidence score
- Improve decision transparency

Output

- Explainable AI Report
-

9. Analytics Agent

Purpose

Generates business analytics for enterprise stakeholders.

Responsibilities

- Calculate business KPIs
- Generate executive metrics
- Summarize enterprise performance

Output

- Analytics Dashboard
-

10. Persona Engine**Purpose**

Generates personalized insights for different enterprise stakeholders.

Supported Personas

- Executive
- Sales
- Compliance
- Technical

Each persona receives customized business insights and recommendations based on its organizational responsibilities.

11. Knowledge Graph Generator**Purpose**

Visualizes relationships extracted from enterprise documents.

Responsibilities

- Construct enterprise knowledge graph
- Display entity relationships
- Improve business understanding
- Support interactive exploration

Output

- Interactive Knowledge Graph
-

12. Memory Agent

Purpose

Maintains shared enterprise context throughout workflow execution.

Responsibilities

- Store workflow context
- Preserve extracted knowledge
- Share information across agents
- Support future workflow extensions

Output

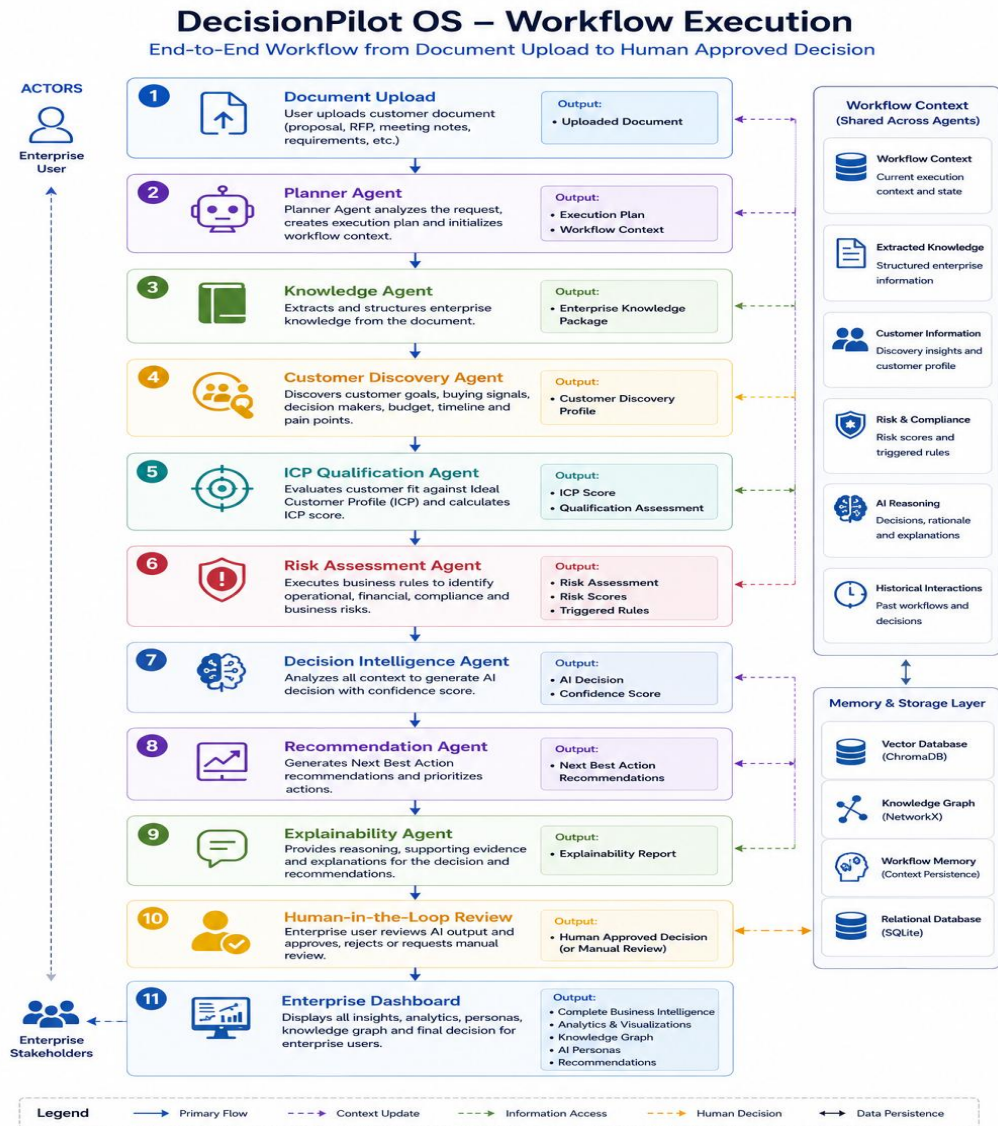
- Shared Enterprise Memory
-

Benefits of the Multi-Agent Architecture

The modular architecture provides several advantages:

- Dynamic planner-based orchestration
- Independent and reusable AI agents
- Shared workflow context
- Improved explainability
- Easy integration of new agents
- Scalable enterprise architecture
- Clear separation of responsibilities
- Support for multiple business domains

This architecture enables DecisionPilot OS to function as a reusable enterprise decision intelligence platform capable of adapting to different business workflows with minimal architectural changes.



6. Workflow Execution

Overview

DecisionPilot OS follows a structured workflow in which enterprise customer documents are processed through a sequence of specialized AI agents. Each agent performs a dedicated task while sharing information through a centralized Workflow Context managed by the Planner Agent.

This workflow ensures that every business recommendation is generated using contextual enterprise knowledge, structured reasoning, explainable AI, and Human-in-the-Loop review before a final decision is accepted.

Workflow Steps

Step 1 – Document Upload

The workflow begins when an enterprise user uploads a customer document through the web dashboard. Supported documents include business proposals, meeting notes, customer requirement documents, RFPs, and enterprise reports.

Output

- Uploaded enterprise document
-

Step 2 – Planner Agent

The Planner Agent receives the request and creates an execution plan. It determines the sequence of specialized AI agents and initializes the shared Workflow Context that is used throughout the execution pipeline.

Output

- Execution Plan
 - Workflow Context
-

Step 3 – Knowledge Extraction

The Knowledge Agent analyzes the uploaded document and extracts structured enterprise information such as company details, industry, stakeholders, business objectives, timelines, budgets, and other relevant business entities.

Output

- Enterprise Knowledge Package
-

Step 4 – Customer Discovery

The Customer Discovery Agent identifies business opportunities and customer intent by extracting:

- Business goals
- Buying signals
- Decision makers
- Budget
- Project timeline

- Customer pain points

Output

- Customer Discovery Profile
-

Step 5 – ICP Qualification

The ICP Qualification Agent evaluates how closely the customer matches the organization's Ideal Customer Profile (ICP) using configurable business criteria.

Output

- ICP Score
 - Customer Qualification
-

Step 6 – Risk Assessment

The Risk Assessment Agent executes configurable business rules to evaluate operational, financial, compliance, and business risks.

The agent generates an overall enterprise risk score together with detailed risk categories and triggered business rules.

Output

- Enterprise Risk Assessment
 - Risk Scores
 - Triggered Rules
-

Step 7 – Decision Intelligence

The Decision Intelligence Agent combines customer qualification, enterprise knowledge, and risk assessment to determine the most appropriate business decision.

Each decision includes a confidence score that represents the AI's confidence in the generated recommendation.

Output

- AI Decision
- Confidence Score

Step 8 – Recommendation Generation

The Recommendation Agent generates actionable Next Best Action recommendations that guide enterprise users on the most appropriate business actions.

Recommendations are prioritized according to business context, opportunity value, and risk level.

Output

- Next Best Action Recommendations

Step 9 – Explainable AI

The Explainability Agent generates transparent reasoning for every recommendation by providing:

- Supporting evidence
- Business reasoning
- Confidence explanation
- Decision justification

This improves trust and transparency for enterprise decision makers.

Output

- Explainability Report

Step 10 – Human-in-the-Loop Review

Before recommendations are accepted, enterprise users review the AI-generated decision through the Human-in-the-Loop interface.

Users can:

- Approve
- Reject
- Request Manual Review

This ensures that final business decisions remain under human control while AI acts as an intelligent decision support system.

Output

- Human Approved Decision
-

Step 11 – Visualization and Analytics

Finally, the platform presents all generated intelligence through an interactive enterprise dashboard, including:

- Executive Dashboard
- AI Decision Intelligence
- Customer Discovery
- Risk Assessment
- Enterprise Knowledge Graph
- AI Personas
- Recommendations
- Enterprise Analytics

The dashboard enables business users to explore enterprise insights, understand AI reasoning, and make informed business decisions.

Workflow Summary

The workflow transforms unstructured enterprise documents into structured business intelligence by combining planner-based orchestration, specialized AI agents, enterprise knowledge retrieval, explainable AI, and Human-in-the-Loop review.

This execution pipeline enables DecisionPilot OS to deliver intelligent, transparent, and reusable Next Best Action recommendations for enterprise decision-making.

7. Tools & Technologies Used

Overview

DecisionPilot OS is built using a modern full-stack technology stack that combines AI frameworks, backend services, frontend technologies, graph visualization, vector storage,

and enterprise data processing. Each technology was selected based on scalability, modularity, performance, and ease of integration within an Agentic AI architecture.

Technology	Purpose
Python	Core programming language used for implementing AI agents, business logic, workflow execution, and backend services.
FastAPI	High-performance REST API framework used for document upload, workflow execution, and communication between the frontend and backend.
React.js	Frontend framework used to build the interactive enterprise dashboard and user interface.
Tailwind CSS	Utility-first CSS framework used for responsive and modern UI design.
LangGraph	Agent orchestration framework used to coordinate the execution of multiple specialized AI agents through a planner-based workflow.
Google Gemini 2.5 Flash	Large Language Model (LLM) used for enterprise knowledge extraction, reasoning, summarization, and recommendation generation.
ChromaDB	Vector database used to store enterprise knowledge and support contextual memory retrieval.
SQLite	Lightweight relational database used for local data storage and workflow persistence.
NetworkX	Graph processing library used to construct and manage enterprise knowledge relationships.
React Flow	Interactive visualization library used to display the Enterprise Knowledge Graph.
Lucide React	Icon library used to provide a clean and consistent enterprise user interface.
Pydantic	Data validation framework used to define structured models and ensure reliable data exchange across agents.
Uvicorn	ASGI server used to deploy and execute the FastAPI backend efficiently during development.

Development Environment

The platform was developed using the following tools:

- Visual Studio Code
 - Python Virtual Environment (venv)
 - Git & GitHub
 - Node.js & npm
 - Google AI Studio (Gemini API)
-

Architecture Characteristics

The selected technology stack enables the platform to provide:

- Modular multi-agent architecture
- High-performance REST APIs
- Dynamic planner-based orchestration
- Enterprise knowledge retrieval
- Explainable AI
- Interactive knowledge graph visualization
- Shared workflow context and memory
- Responsive enterprise dashboard
- Human-in-the-Loop decision support

The combination of these technologies provides a scalable and reusable foundation for building intelligent enterprise decision-making platforms across multiple business domains.

8. Platform Features

Overview

DecisionPilot OS provides a comprehensive set of enterprise AI capabilities that automate customer analysis, business reasoning, and decision intelligence through a reusable multi-agent architecture. Each feature is designed to support enterprise users in making faster, more accurate, and explainable business decisions.

Key Features

Dynamic Planner-Based Agent Orchestration

The platform uses a Planner Agent to dynamically coordinate specialized AI agents and manage the complete execution workflow. This modular orchestration enables reusable and extensible enterprise workflows.

Multi-Agent Decision Intelligence

DecisionPilot OS decomposes complex business problems into specialized tasks executed by dedicated AI agents. Each agent performs an independent responsibility while sharing contextual information through a centralized workflow context.

Enterprise Knowledge Extraction

The platform automatically extracts structured business information from uploaded customer documents, including company information, decision makers, business objectives, budgets, timelines, and enterprise requirements.

Customer Discovery Intelligence

Automatically identifies customer goals, buying signals, pain points, executive sponsors, business priorities, and project requirements to build a comprehensive customer profile.

Ideal Customer Profile (ICP) Qualification

Evaluates customer suitability using configurable qualification criteria and generates an ICP score to support business prioritization.

Enterprise Risk Assessment

Performs rule-based business risk analysis by evaluating operational, financial, business, and compliance risks while generating an overall enterprise risk score.

AI Decision Intelligence

Generates intelligent business decisions with confidence scores by combining customer qualification, enterprise knowledge, and risk analysis.

Next Best Action Recommendations

Produces prioritized and actionable recommendations that guide enterprise users toward the most appropriate business actions based on AI-generated insights.

Explainable AI

Provides transparent reasoning for every recommendation through supporting evidence, business justification, confidence scores, and decision explanations, improving trust and accountability.

Interactive Enterprise Knowledge Graph

Automatically constructs an interactive knowledge graph that visualizes relationships between enterprise entities such as organizations, stakeholders, goals, risks, business signals, and recommendations.

AI Persona Intelligence

Generates customized business insights for different enterprise stakeholders, including Executive, Sales, Compliance, and Technical personas, enabling role-specific decision support.

Enterprise Analytics Dashboard

Presents business intelligence through an interactive dashboard featuring executive metrics, workflow analytics, customer insights, risk visualization, AI recommendations, and performance indicators.

Human-in-the-Loop Review

Supports enterprise governance by allowing business users to review AI-generated recommendations before they are accepted or executed. Users can approve, reject, or request manual review of AI decisions.

Shared Workflow Context and Memory

Maintains shared contextual information throughout workflow execution, allowing all AI agents to access consistent enterprise knowledge and ensuring coordinated decision-making across the platform.

Platform Benefits

DecisionPilot OS delivers the following benefits to enterprise organizations:

- Faster enterprise decision-making
- Reduced manual analysis effort
- Improved customer qualification
- Explainable and transparent AI recommendations
- Modular and reusable AI architecture
- Scalable enterprise workflows
- Enhanced business visibility
- Human-centered AI governance

These capabilities collectively enable organizations to transform enterprise customer interactions into intelligent, explainable, and actionable business decisions using a reusable Agentic AI platform.

9. Future Enhancements

Overview

DecisionPilot OS has been designed as a reusable and extensible Agentic Decision Intelligence Platform. Its modular architecture allows new AI agents, enterprise integrations, business rules, and workflows to be incorporated with minimal changes to the existing system.

The following enhancements are planned for future versions of the platform.

Planned Enhancements

Enterprise CRM Integration

Integrate with enterprise CRM platforms such as Salesforce, Microsoft Dynamics 365, and HubSpot to automatically retrieve customer interactions, opportunity data, account history, and sales activities.

Enterprise Communication Integration

Connect with enterprise communication platforms such as Microsoft Teams, Slack, Outlook, and Gmail to automatically analyze meeting transcripts, emails, and customer conversations.

Advanced Document Processing

Extend document ingestion to support multiple enterprise formats, including PDF, DOCX, PPTX, spreadsheets, contracts, proposals, meeting transcripts, and scanned documents with OCR capabilities.

Scenario Simulation Agent

Introduce a Scenario Intelligence Agent capable of performing "What-If" business analysis by simulating changes in budgets, stakeholders, compliance requirements, and business risks to evaluate alternative decision outcomes.

Continuous Learning Intelligence

Enhance the platform with a Learning Intelligence Agent that continuously improves recommendations by analyzing historical decisions, user feedback, and previously approved or rejected business actions.

Multi-Domain Support

Extend the reusable platform beyond Enterprise SaaS Sales to support additional business domains such as Customer Success, Banking, Healthcare, Insurance, Manufacturing, Staffing, Energy, and Supply Chain Management.

Enterprise Authentication and Role-Based Access

Implement secure authentication and role-based access control (RBAC) to provide personalized dashboards and controlled access for executives, sales teams, compliance officers, and technical stakeholders.

Cloud Deployment

Deploy the platform on cloud infrastructure with support for containerized deployment, scalable microservices, centralized monitoring, and enterprise-grade availability.

Vision

DecisionPilot OS is designed as a long-term enterprise AI platform capable of evolving beyond a single business use case. By leveraging reusable AI agents, planner-based orchestration, shared memory, explainable AI, and modular workflows, the platform can be adapted to support diverse enterprise decision-making scenarios while maintaining scalability, transparency, and extensibility.

10. Setup Instructions

System Requirements

Before running DecisionPilot OS, ensure the following software is installed:

- Python 3.11 or later
 - Node.js 18 or later
 - npm
 - Git
 - Google Gemini API Key
-

Installation

Step 1 – Clone the Repository

```
git clone <repository-url>
```

```
cd DecisionPilot-OS
```

Step 2 – Backend Setup

Navigate to the backend directory.

```
cd backend
```

Create a virtual environment.

```
python -m venv venv
```

Activate the virtual environment.

Windows

```
venv\Scripts\activate
```

Linux / macOS

```
source venv/bin/activate
```

Install the required dependencies.

```
pip install -r requirements.txt
```

Configure the environment variables by creating a .env file.

Example:

```
GOOGLE_API_KEY=YOUR_GEMINI_API_KEY
```

Start the FastAPI server.

```
uvicorn main:app --reload
```

The backend will be available at:

```
http://127.0.0.1:8000
```

Step 3 – Frontend Setup

Open a new terminal.

Navigate to the frontend directory.

```
cd frontend
```

Install dependencies.

```
npm install
```

Start the React application.

```
npm run dev
```

The frontend will be available at:

```
http://localhost:5173
```

Running the Platform

1. Launch the FastAPI backend.
 2. Launch the React frontend.
 3. Open the application in your browser.
 4. Upload an enterprise customer document.
 5. The Planner Agent orchestrates the workflow.
 6. The dashboard displays enterprise insights, recommendations, analytics, AI personas, the knowledge graph, and explainable AI results.
 7. Review the AI recommendation using the Human-in-the-Loop interface.
-

Project Repository

The source code includes:

- Backend implementation
 - Frontend application
 - AI agents
 - Business engines
 - Planner orchestration
 - Enterprise memory
 - Knowledge graph generation
 - Documentation
 - Setup instructions
-

Conclusion

DecisionPilot OS demonstrates a reusable Agentic Decision Intelligence Platform that combines dynamic planner-based orchestration, specialized AI agents, enterprise knowledge retrieval, explainable AI, Human-in-the-Loop review, and interactive analytics to generate intelligent Next Best Action recommendations. The modular architecture enables the platform to be extended to multiple enterprise domains while maintaining scalability, transparency, and reusability.